Lecture 3: Planetary Heat & Energy Transport

Introduction to Planetary Science

Tim Lichtenberg

Kapteyn Astronomical Institute
University of Groningen

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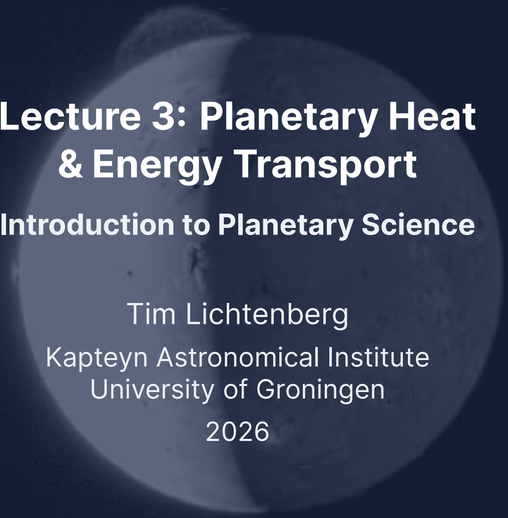


Io volcanic plume, New Horizons (2007). Credit: NASA/JHUAPL/SwRI

Learning Objectives

By the end of this lecture, you will be able to:

- Identify the main energy sources driving planetary evolution
- Distinguish conduction, convection, and radiation as heat transport mechanisms
- Derive and apply the conductive cooling timescale $\tau \sim L^2/\kappa$
- Explain the Rayleigh and Nusselt numbers and what they tell us
- Describe tidal heating and why Io is the most volcanic body in the solar system

Recap: Lecture 2

- Protoplanetary disks: structure, snow line, 3–5 Myr lifetime
- Dust → planetesimals → planets (streaming instability, pebble accretion)
- Core accretion vs. gravitational instability for giant planets
- Kepler's laws, vis-viva equation
- Orbital resonances, tidal forces, migration

Today: What happens *inside* planets after they form?

1. Energy Sources

Io’s Tvashtar plume, New Horizons (2007). Credit: NASA/JHUAPL/SwRI

What Heats a Planet?

Four principal sources of internal heat:

1. **Accretional heating:** gravitational energy from assembly
2. **Gravitational differentiation:** core formation releases ~10% more
3. **Radioactive decay:** sustained for billions of years
4. **Tidal heating:** orbital energy dissipated as friction

Without internal heat → no volcanism, no magnetic field, no plate tectonics, no atmospheric outgassing.

Accretional Heating

Gravitational energy of assembling a uniform sphere:

$$E_{\mathrm{acc}} \sim \frac{3}{5}\frac{GM^2}{R}$$

For Earth: $E_{\mathrm{acc}} \approx 2.2 \times 10^{32}$ J

If all retained as heat:

$$\Delta T = \frac{E_{\mathrm{acc}}}{Mc_p} \approx \frac{2.2 \times 10^{32}}{5.97 \times 10^{24} \times 1200} \approx 30{,}000\ \mathrm{K}$$

Enough to melt Earth many times over. Accretion inevitably produces a **magma ocean**.

Where Does the Accretional Energy Go?

Not all E_{acc} is retained as interior heat.

- **Radiation to space:** hot surface cools efficiently during late accretion
- **Thermal blanketing:** a thick primordial atmosphere traps heat, raising retention
- **Depth of deposition:** energy from giant impacts is buried deep, while small impacts deposit near the surface
- Typical retained fraction: 10–50% (depends on impact size distribution)

Even 10% retention is enough to fully melt a terrestrial planet: the magma ocean is effectively guaranteed for bodies $\gtrsim$ Mars-sized.

Where Do Heat-Producing Elements Live?

Preview for Lecture 4: elements partition differently between metal and silicate phases.

- **Lithophile** ("rock-loving"): partition into silicate phase
  - U, Th, K: the long-lived heat producers ⇒ **mantle and crust**
  - Si, Mg, Al, Ca, Fe oxides (most of the rock-forming elements)
- **Siderophile** ("iron-loving"): partition into metallic iron
  - Fe, Ni, Co, Pt, Ir, Au ⇒ **core**
  - Very little radiogenic heating in the core today

Because U, Th, K are lithophile, radiogenic heat is produced in the **mantle**, not the core. This controls the direction of heat flow and sustains mantle convection.

Source: Goldschmidt (1937); McDonough (2003)

Gravitational Differentiation

A second, one-time burst of heat accompanies **core formation**.

- Dense iron droplets sink from the magma ocean to the centre
- Gravitational potential energy → heat via viscous friction
- Energy release ~10% of the accretional budget

Order-of-magnitude estimate for Earth:

$$E_{\mathrm{diff}} \sim \frac{3}{10}\,\frac{GM_{\mathrm{core}}^2}{R_{\mathrm{core}}}\left(\frac{R}{R_{\mathrm{core}}} - 1\right) \approx 2 \times 10^{31}\ \mathrm{J}$$

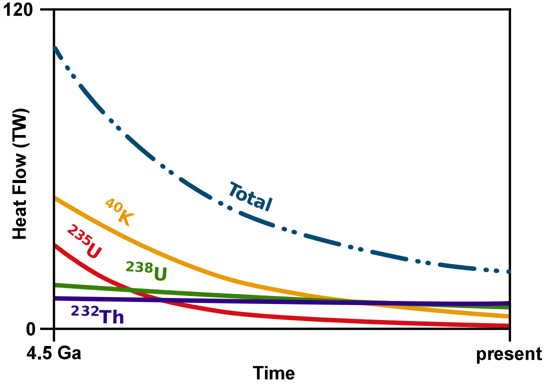
This heat is released **once**, during core formation, and sustains the magma ocean long enough for chemical differentiation (Lecture 4).

Radioactive Decay: The Long-Lived Engine

Isotope	Half-life (Gyr)	Heat ($\mathrm{W\,kg^{-1}}$)	Concentration
$^{238}\mathrm{U}$	4.47	9.5×10^{-5}	20 ppb
$^{235}\mathrm{U}$	0.704	5.7×10^{-4}	0.14 ppb
$^{232}\mathrm{Th}$	14.0	2.6×10^{-5}	80 ppb
$^{40}\mathrm{K}$	1.25	2.9×10^{-5}	240 ppb

- Lithophile elements → concentrate in mantle/crust
- Present-day radiogenic heat: ~20 TW
- 4.5 Gyr ago: 3–4× higher (short-lived isotopes more abundant)

Radiogenic Heat Over Time



Total heat production (black) is the sum of four isotopes. $^{40}\mathrm{K}$ and $^{235}\mathrm{U}$ dominated in the early solar system.

Short-Lived Isotopes: $^{26}\mathrm{Al}$

- Half-life: only 0.72 Myr (now extinct)
- Produced by nearby stellar nucleosynthesis, delivered into the protoplanetary disk
- Initial $^{26}\mathrm{Al}/^{27}\mathrm{Al}$ ratio ~ 5×10^{-5} (canonical)
- In the first few Myr: extraordinarily potent heat source (10^{-7} W kg^{-1} initially)
- Could melt planetesimals as small as a few km radius
- Initiated core formation and volatile loss **before** planets fully assembled

$^{26}\mathrm{Al}$ is why even small asteroids show evidence of melting and differentiation in the meteorite record.

Where Did $^{26}\mathrm{Al}$ Come From?

Short-lived isotopes require a nearby stellar source just before solar system formation.

- $^{26}\mathrm{Al}$ is produced in:
  - Massive star winds (Wolf-Rayet stars)
  - Core-collapse supernovae
- With a 0.72 Myr half-life, it must reach the nascent solar nebula within ~1–2 Myr of production
- Implies the Sun formed in a **massive star cluster** or near an active star-forming region
- $^{60}\mathrm{Fe}$: produced primarily by supernovae; detection confirms contribution from a nearby SN
- Isotopic abundances in meteorites constrain: one nearby massive star, or multiple injection events
- The Sun's birth environment may have been an Orion-like star-forming cluster

Other Short-Lived Radionuclides

The young solar system contained a full zoo of short-lived isotopes:

Isotope	Half-life (Myr)	Daughter	Role
$^{26}\mathrm{Al}$	0.72	$^{26}\mathrm{Mg}$	Dominant heating
$^{60}\mathrm{Fe}$	2.6	$^{60}\mathrm{Ni}$	Minor heating; debated initial abundance
$^{53}\mathrm{Mn}$	3.7	$^{53}\mathrm{Cr}$	Chronometer (not major heater)
$^{182}\mathrm{Hf}$	8.9	$^{182}\mathrm{W}$	Core formation chronometer (Lecture 4)
$^{129}\mathrm{I}$	16	$^{129}\mathrm{Xe}$	Atmosphere formation timing

The production of these isotopes likely required injection from a nearby supernova or AGB star into the molecular cloud that formed the solar system.

Extinct Radionuclide Chronometers

Short-lived isotopes don't just heat planetesimals; they also act as **clocks** for early differentiation.

System	Half-life	Daughter	What it dates
$^{182}\mathrm{Hf}$–$^{182}\mathrm{W}$	8.9 Myr	W	Core formation
$^{26}\mathrm{Al}$–$^{26}\mathrm{Mg}$	0.72 Myr	Mg	Planetesimal crystallisation
$^{53}\mathrm{Mn}$–$^{53}\mathrm{Cr}$	3.7 Myr	Cr	Early differentiation events
$^{129}\mathrm{I}$–$^{129}\mathrm{Xe}$	16 Myr	Xe	Atmosphere retention / volatile loss
$^{244}\mathrm{Pu}$–Xe fission	81 Myr	Xe	Closure of mantle degassing

Combined, these systems pin the first ~100 Myr of solar system history to within a few Myr.

Source: Kleine & Wadhwa (2017), *Rev. Mineral. Geochem.*

How Heat Production Has Changed Over Time

- t = 0 (solar system formation): $^{26}\mathrm{Al}$ dominates (~10^{-7} W kg^{-1})
- t = 5 Myr: $^{26}\mathrm{Al}$ mostly gone
- t = 10–100 Myr: transition to long-lived isotopes
- t = 0.5 Gyr: $^{235}\mathrm{U}$ and $^{40}\mathrm{K}$ dominate
- t = 4.5 Gyr (today): $^{238}\mathrm{U}$ and $^{232}\mathrm{Th}$ dominate
- Total radiogenic heat today is ~25–30% of its value at formation

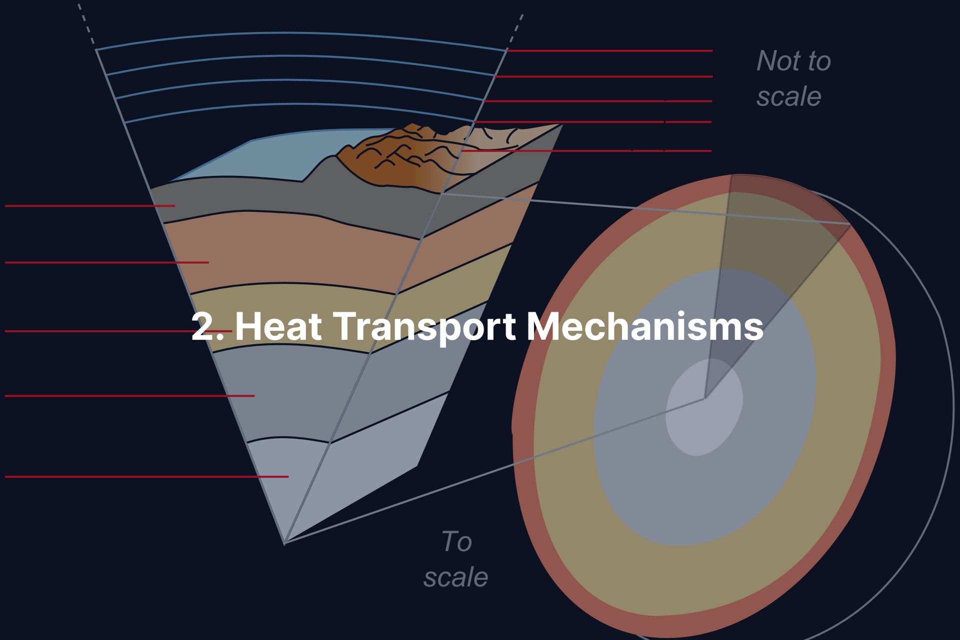
Small bodies melted early thanks to $^{26}\mathrm{Al}$, even if their long-lived budget was modest.

Earth's Heat Budget: 47 TW

Source	Contribution
Radiogenic heating (mantle + crust)	~20 TW
Primordial (secular cooling)	~15 TW
Core cooling and solidification	~10 TW
Tidal dissipation (from the Moon)	~0.1 TW
Total	**~47 TW**

~40% from ongoing radioactive decay; 60% from primordial heat.
Earth is still cooling.

2. Heat Transport Mechanisms



Earth’s interior. Credit: Wikimedia Commons, CC BY-SA 3.0

Three Ways to Move Heat

Conduction: Heat transferred by molecular/atomic collisions
$\vec{q} = -k\nabla T$ (Fourier's law)
Slow, diffusive; dominant in lithosphere, small bodies

Convection: Heat transported by bulk fluid motion
Hot material rises, cool material sinks
Dominant in mantles, outer cores, giant planet interiors

Radiation: Energy carried by electromagnetic waves
$F = \epsilon\sigma T^4$ (Stefan–Boltzmann)
Dominant at surfaces and in atmospheres

Conduction: Fourier's Law

- Heat flux: $\vec{q} = -k\nabla T$
- Thermal diffusivity: $\kappa = k/(\rho c_p) \approx 10^{-6}$ m^2 s^{-1} (rock)
- Inherently **diffusive**, slow over large distances
- The time for heat to conduct through distance L:

$$\tau_{\mathrm{cond}} \sim \frac{L^2}{\kappa}$$

We will derive this next.

Thermal Conductivity: Typical Values

Material	k (W m^{-1} K^{-1})	κ (m^2 s^{-1})
Silicate rock (upper mantle)	3–5	$\sim 10^{-6}$
Bridgmanite (lower mantle)	5–10	$\sim 2\times10^{-6}$
Water ice	2.2	1.1×10^{-6}
Iron (solid)	80	$\sim 2\times10^{-5}$
Air (atmosphere)	0.025	$\sim 2\times10^{-5}$

- Metals have high k but also high ρc_p: intermediate κ
- Rocks have low k and low ρc_p: similar κ
- This is why the L^2/κ scaling matters more than k alone

When Does Radiation Dominate?

The Stefan–Boltzmann law $F = \epsilon\sigma T^4$ gives the heat flux from a surface at temperature T.

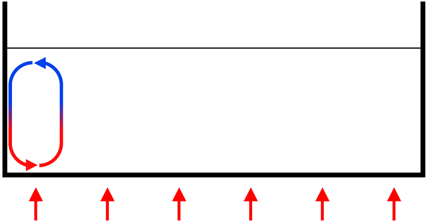
Radiation matters in:

- **Planetary surfaces:** heat loss to space (Lecture 5 energy balance)
- **Optically thin atmospheres:** e.g. Mars' upper atmosphere
- **Magma oceans:** molten silicate at $T \gtrsim 2000$ K radiates efficiently

Why not in solid interiors?

- Silicate rocks are opaque at infrared wavelengths
- Photon mean free paths are much smaller than conductive length scales
- Exception: partially transparent silicates above ~2000 K become radiatively conductive

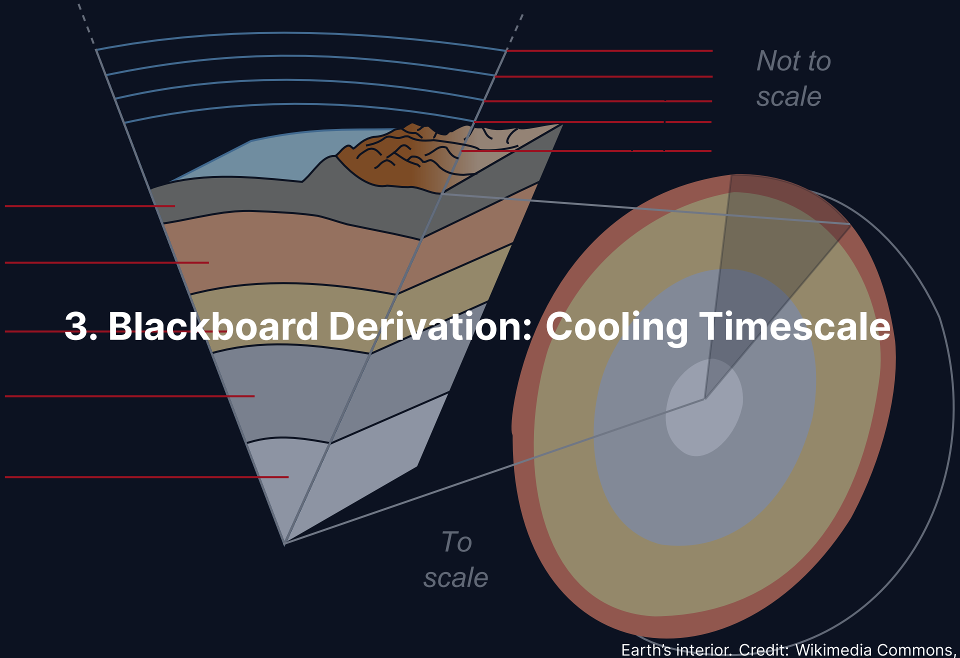
Convection: Nature's Heat Elevator



Credit: Wikimedia Commons, CC BY-SA 3.0

- Hot material at base: buoyant, rises
- Cool material at top: dense, sinks
- Far more efficient than conduction
- Active in:
  - Earth's mantle (~cm/yr)
  - Outer core (liquid metal)
  - Giant planet interiors
  - Atmospheres

3. Blackboard Derivation: Cooling Timescale



Earth's interior. Credit: Wikimedia Commons, CC BY-SA 3.0

Goal and Strategy

Goal: Estimate how long it takes heat to conduct out of a body of size L.

Strategy: Start from the heat equation, then use dimensional analysis instead of solving the PDE.

- We want a characteristic timescale $\tau_{\mathrm{cond}}(L, \kappa)$
- Plug in representative values for rock
- Compare to the age of the solar system (4.6 Gyr)

Physical intuition: bigger bodies have a longer path for heat to diffuse, so they stay hot for longer. We will derive exactly *how much* longer.

Step 1: The Heat Diffusion Equation

Fourier's law $\vec{q} = -k\nabla T$ combined with energy conservation gives:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T)$$

For constant thermal conductivity k, define $\kappa = k/(\rho c_p)$:

$$\frac{\partial T}{\partial t} = \kappa \nabla^2 T$$

In one dimension:

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2}$$

The rate of temperature change is proportional to the **curvature** of the temperature profile; flat profiles do not evolve, curved ones relax.

Step 2: Dimensional Analysis

Replace derivatives with characteristic scales:

- Time derivative: $\frac{\partial T}{\partial t} \sim \frac{T}{\tau}$
- Spatial derivative: $\frac{\partial^2 T}{\partial x^2} \sim \frac{T}{L^2}$

Substitute into the heat equation:

$$\frac{T}{\tau} \sim \kappa \frac{T}{L^2}$$

Temperature cancels (as it must: the timescale is intrinsic, not tied to any particular T):

$$\boxed{\tau_{\text{cond}} \sim \frac{L^2}{\kappa}}$$

Cooling Timescales Across the Solar System

Using $\kappa \approx 10^{-6}$ m^2 s^{-1} (silicate rock):

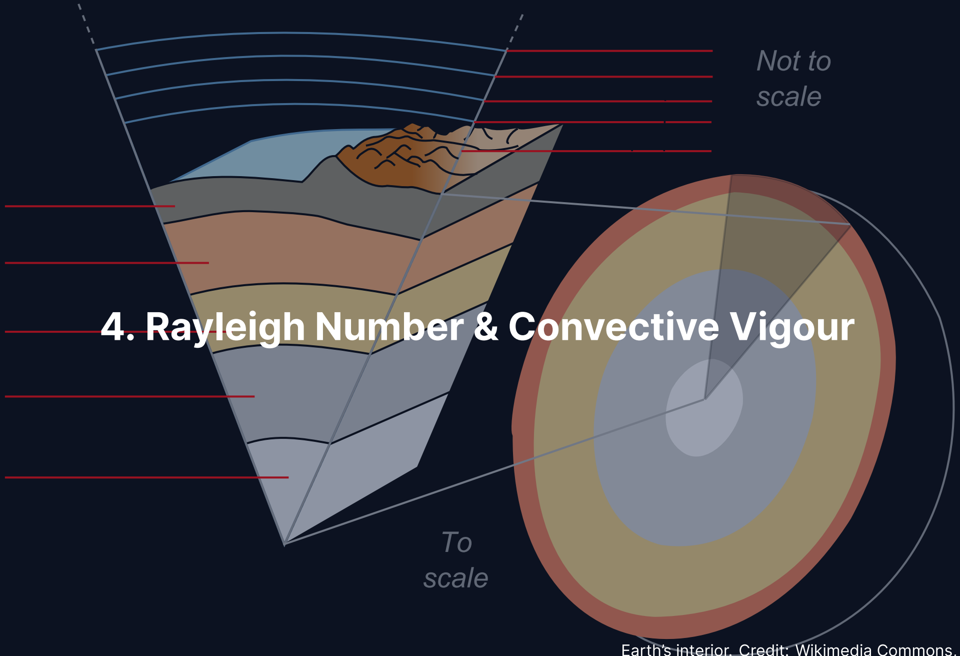
Body	Size L	$\tau = L^2/\kappa$	Familiar units
1 m boulder	1 m	10^6 s	~12 days
100 km asteroid	10^5 m	10^{16} s	~300 Myr
Moon	1.7×10^6 m	3×10^{18} s	~100 Gyr
Earth	6.4×10^6 m	4×10^{19} s	~1300 Gyr

Earth's conductive cooling time $\gg$ age of the universe.
Yet Earth loses 47 TW $\Rightarrow$ there **must** be convection.

Break

Earthrise, Apollo 8 (1968). Credit: NASA

4. Rayleigh Number & Convective Vigour



Earth’s interior. Credit: Wikimedia Commons, CC BY-SA 3.0

The Rayleigh Number

Does a fluid layer convect or merely conduct?

$$\mathrm{Ra} = \frac{\alpha \rho g \Delta T\, d^3}{\kappa \eta}$$

Numerator (buoyancy):

- α = thermal expansion
- ρ = density
- g = gravity
- ΔT = temperature contrast
- d = layer depth

Denominator (dissipation):

- κ = thermal diffusivity
- η = dynamic viscosity

Critical value: $\mathrm{Ra}_c \approx 1000$

Earth's Mantle: Vigorous Convection

For Earth's mantle:

$$\mathrm{Ra}_\oplus \sim \frac{2 \times 10^{-5} \times 4000 \times 10 \times 2500 \times (3 \times 10^6)^3}{10^{-6} \times 10^{21}} \sim 10^7\text{–}10^8$$

- $\mathrm{Ra}_\oplus \gg \mathrm{Ra}_c \approx 1000$: **vigorous convection confirmed**
- Explains why Earth's interior cools efficiently despite conduction being inadequate

The Nusselt number $\mathrm{Nu} \propto \mathrm{Ra}^{1/3}$ measures convection efficiency.
For Earth: $\mathrm{Nu} \sim (10^7)^{1/3} \sim 200$, so convection transports heat 200× faster than conduction.

Rayleigh Number Across the Solar System

Body	Ra	Regime
Earth's mantle	10^7–10^8	Vigorous convection
Venus' mantle	10^7–10^8	Vigorous (stagnant lid)
Mars' mantle	10^6–10^7	Moderate convection
Moon's mantle	10^4–10^5	Weak; cooling by conduction
Mercury's mantle	$\sim 10^4$	Marginal; largely cooled
Magma ocean	$\gtrsim 10^{25}$	Turbulent convection
Earth's outer core	$\gtrsim 10^{25}$	Turbulent (dynamo)

Ra is hyper-sensitive to size ($\propto d^3$) and viscosity ($\propto 1/\eta$). Small bodies and very viscous interiors quickly drop below Ra_c.

Nusselt Number: Efficiency of Convection

- $\mathrm{Nu} = q_{\mathrm{total}}/q_{\mathrm{cond}}$ = actual flux over conductive flux
- Nu = 1: pure conduction
- Nu ≫ 1: vigorous convection

Classical scaling from boundary-layer theory:

$$\mathrm{Nu} \propto \mathrm{Ra}^{\beta}, \quad \beta \approx 1/3$$

Consequences:

- Doubling Ra only ~26% more efficient (weak sensitivity)
- Increasing Ra by 10^3: Nu increases by 10×
- Earth with Ra ~ 10^7 has Nu ~ 200

Source: Schubert et al. (2001), *Mantle Convection in the Earth and Planets*

Thermal Boundary Layers

A convecting interior develops three layers:

1. **Cold boundary layer (top):** conductive, steep *T* drop
   → Earth's **lithosphere** (~100–200 km)
2. **Well-mixed interior:** nearly adiabatic, slow *T* variation
   → bulk of the mantle
3. **Hot boundary layer (bottom):** steep *T* rise into the core
   → Earth's **D" layer** (~200 km)

Nearly all the temperature drop occurs in the two thin boundary layers.

The Adiabatic Gradient

In the well-mixed convecting interior, temperature follows an **adiabat**:

$$\left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{\mathrm{adiabat}} = -\frac{\alpha g T}{c_p}$$

- α = thermal expansion, g = gravity, c_p = specific heat
- For Earth's upper mantle: ~0.3 K/km (a few K per km)
- Much shallower than the conductive gradient in the lithosphere (~25 K/km)
- Adiabatic profiles are nearly isothermal on the scale of the mantle

Temperature changes only a few hundred K over the entire convecting mantle: efficient mixing keeps the interior nearly adiabatic.

Two Tectonic Regimes

Mobile lid (plate tectonics):

- Cold boundary layer **breaks** into plates
- Plates subduct, recycle material
- Most efficient heat loss
- **Only Earth**

Stagnant lid:

- Single thick, immobile shell
- Heat escapes by conduction + volcanism
- Less efficient cooling
- Venus, Mars, Mercury, Moon

Why Earth is unique: likely a combination of water, size, and thermal history.

Why Does Earth Have Plate Tectonics?

One of the biggest open questions in planetary geophysics.

Proposed factors:

- **Water:** weakens rocks at fault zones; lowers melting point in subducting slabs
- **Planet size:** higher convective stresses in a larger mantle
- **Thermal state:** must be cold enough for a rigid lid but hot enough to sustain plume-driven rifting
- **Prior history:** once plate tectonics starts, it may be self-sustaining (recycling weakens the lid)
- Venus may have had plate tectonics in the past (tessera highlands); lost it after drying out

Habitability-oriented view: plate tectonics is linked to liquid water, and both are linked to long-term climate stability.

Heat-Pipe Volcanism

An alternative heat-loss regime for very hot bodies.

- Magma rises through conduits (“heat pipes”)
- Erupts on the surface, cools, and is buried by new lava
- Old crust subsides into the hot interior as new lava covers it
- **Io today:** global resurfacing by volcanism every ~10^6 yr
- **Early Earth:** may have been a heat-pipe regime (Hadean, >4 Ga)
- Allows efficient cooling of a very hot mantle without plate tectonics

Source: Turcotte (1989); Moore & Webb (2013), *Nature*

Mantle Plumes and Hotspots

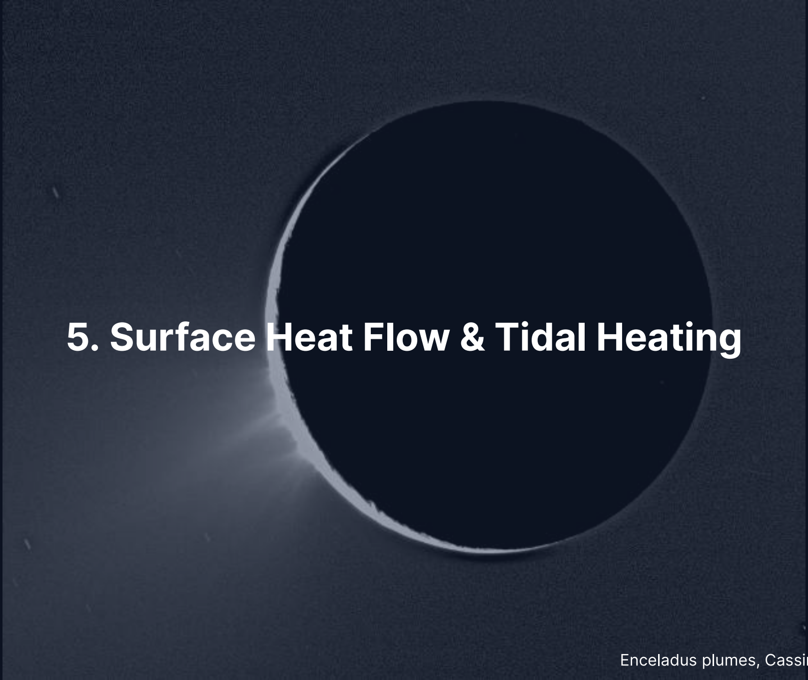
Whole-mantle convection is dominated by sheet-like downwellings (subducting slabs), but the hot bottom boundary layer is also unstable.

- Hot, buoyant material collects at the base of the mantle (D″)
- Periodically detaches as narrow, rising **plumes** (~100 km diameter)
- Reach the lithosphere and generate **intraplate volcanism**
- Plumes are long-lived (10^7–10^8 yr) and approximately fixed in the deep mantle

Examples (Earth): Hawaii, Iceland, Yellowstone, Réunion.

Plumes move heat from the core–mantle boundary to the surface independently of plate tectonics.

5. Surface Heat Flow & Tidal Heating



Enceladus plumes, Cassini. Credit: NASA/JPL/SSI

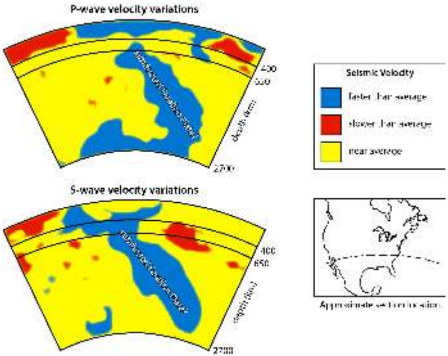
Core Cooling and the Dynamo

The mantle acts as a thermostat on the core.

- Heat flux from core to mantle sets the rate of core cooling
- A vigorously convecting mantle extracts heat efficiently, enabling core convection
- Core cooling drives the inner core's gradual crystallisation
- Inner core growth releases latent heat + light elements → compositional buoyancy
- This compositional convection is the **dominant** driver of today's geodynamo (Lecture 4)

Earth's magnetic field ultimately exists because mantle convection efficiently cools the core; stagnant-lid Venus likely lacks this.

Whole-Mantle vs. Layered Convection



Credit: Wikimedia (USGS/Caltech), CC BY-SA

The debate: does Earth's mantle convect as one layer or two?

- Seismic tomography images the Farallon slab descending to the core–mantle boundary
- Plumes rise from the hot bottom boundary layer through the full mantle
- Strong evidence for **whole-mantle convection**
- Partial barrier at 660 km (phase transition) slows, but does not stop, slabs
- Modern consensus: mantle is one

Plumes vs. Plates: Two Modes of Upwelling

- **Mid-ocean ridges:** passive upwelling where plates pull apart
- Temperatures close to the ambient mantle
- **Mantle plumes:** active upwelling from the deep mantle
- ~200–300 K hotter than the ambient mantle (from olivine thermobarometry)
- Plumes deliver primordial (3He-rich) material from the lower mantle
- Large Igneous Provinces (LIPs) may be plume-heads: Deccan Traps, Siberian Traps, Ontong-Java Plateau
- LIPs correlate with mass extinction events in geological history

Source: Courtillot & Renne (2003), *Comptes Rendus Geoscience*

Surface Heat Flow

Body	Heat flux (mW m^{-2})	Total (TW)	Source
Earth	~90	~47	Radiogenic + primordial
Moon	10–15	~0.3	Primordial (cooled)
Mars	15–25	~3–5	Radiogenic + primordial
Io	2000–3000	~130–160	**Tidal dissipation**

Source: Davies & Davies (2010); Veeder et al. (2012); InSight + gravity/topography estimates

Io's heat flux is 20–30× higher than Earth's per unit area. Total power (~1.5× Earth's 47 TW) is smaller than the flux ratio suggests because Io's mass is only ~1% of Earth's.

Spatial Pattern of Earth's Heat Flow

Earth's 47 TW is not uniformly distributed:

- **Oceanic regions** (~60% of area): 70% of total heat loss
  - Thin lithosphere, new crust at ridges
  - Flux decreases with crustal age (Stein & Stein 1992 cooling model)
- **Continental regions** (~40% of area): 30% of total heat loss
  - Thick lithosphere insulates the mantle
  - BUT concentration of U, Th, K in the crust adds local contribution
  - ~40% of the continental surface flux is crustal-radiogenic

Source: Jaupart & Mareschal (2011); Davies & Davies (2010)

Tidal Heating: How It Works

- Circular orbit + tidal lock → static bulge, **no heating**
- **Eccentric orbit:** distance varies each orbit
  - Closer: tidal force stronger, moon moves faster
  - Farther: tidal force weaker, moon moves slower
- Continuously varying deformation → **flexing dissipates heat**

Heating depends on:

- Eccentricity e (zero e → zero heating)
- Distance a (heating $\propto a^{-6}$)
- Internal dissipation (tidal quality factor Q)

Secular Cooling of Planets

Planets are thermally non-stationary: they started hot and are gradually cooling.

- Stored heat ~ $E_{\mathrm{acc}} + E_{\mathrm{diff}} + \int L_{\mathrm{rad}}(t)\,dt$
- Lost heat ~ $\int L_{\mathrm{surface}}(t)\,dt$
- The difference is the present-day thermal state
- Earth: still cooling at ~50–100 K/Gyr
- Smaller bodies cool faster (lower thermal inertia, thinner conductive boundary layer)

All terrestrial planets are in net secular cooling; Earth's is slowed by convective resurfacing but not stopped.

Tidal Dissipation Rate

For a synchronously rotating body on an eccentric orbit, the tidal heating power is (to leading order in e):

$$\dot{E}_{\text{tidal}} = \frac{21}{2} \frac{k_2}{Q} \frac{G M_*^2 R^5 n e^2}{a^6}$$

where:

- k_2 = Love number (dimensionless, ~0.3 for rocky bodies)
- Q = tidal quality factor (lower = more dissipative)
- M_* = mass of the primary (Jupiter, Saturn, etc.)
- R = radius of the moon
- n = mean motion (= $\sqrt{GM_*/a^3}$)
- a = semi-major axis, e = eccentricity

Note the steep R^5/a^6 dependence: heating is dramatic for close-in moons

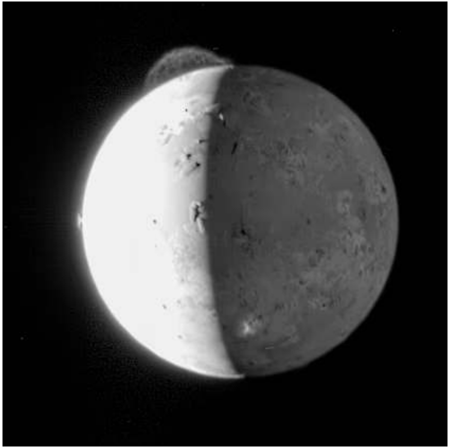
The Laplace Resonance: Sustaining Eccentricity

Io's orbit would circularise in ~10 Myr if left alone. Why is its eccentricity maintained?

- Io–Europa–Ganymede are locked in a 4:2:1 **mean-motion resonance**
- Every time Io laps Europa, the two meet at the same orbital phase
- Periodic gravitational kicks continually **pump** Io's eccentricity
- Without this, tidal dissipation would damp e and shut off the heating
- The resonance was first identified by Laplace in 1805

The three-body resonance is what makes Io's tidal heating indefinite, not transient.

Io: The Most Volcanic Body

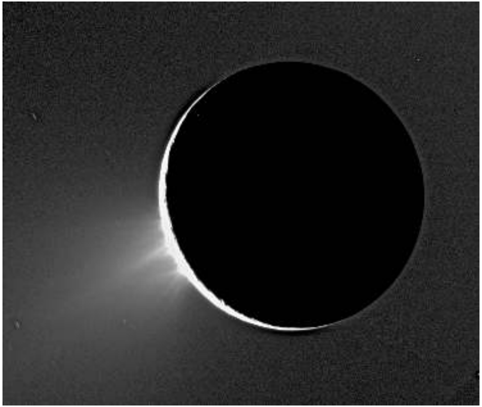


Credit: NASA/JHUAPL/SwRI (2007)

- ~ 10^{14} W of tidal heat (2× Earth's total!)
- Predicted by Peale, Cassen & Reynolds (1979)
- Confirmed days later by Voyager 1
- Powered by the **Laplace resonance**
- >400 active volcanic centres
- Entire surface resurfaced on ~Myr timescale

The Laplace resonance continuously pumps Io's eccentricity, maintaining

Enceladus: A Tiny Moon with a Big Secret



Credit: NASA/JPL/SSI (Cassini)

- Radius only 252 km
- Water ice jets from "tiger stripe" fractures
- Thermal power: 5–15 GW (tidal dissipation)
- Global subsurface ocean beneath ~20–30 km ice
- Plumes contain: salts, silica, organics, H_2
- Hydrothermal activity on the ocean floor

Enceladus demonstrates that tidal heating can sustain liquid water oceans far from the habitable zone.

Where in Enceladus Is Tidal Heat Dissipated?

Two competing scenarios explain the observed 5–15 GW of plume power:

Rocky core dissipation:

- Tidal flexing of porous silicate core
- Heat conducted up through the ocean
- Matches hydrothermal signatures (silica nanoparticles at ~90°C)
- Difficulty: plume power exceeds simple core-heating estimates

Ice-shell dissipation:

- Viscous dissipation concentrated in the warm ice shell base
- Positive feedback: softer ice → more dissipation
- Can reach 10–20 GW with realistic rheology
- Concentrates heat near the south pole tiger stripes

Likely: both contribute, with the ice shell dominating the plume heat flux.

Source: Čadek et al. (2016); Hemingway & Mittal (2019)

Enceladus Plumes: A Habitability Laboratory

Cassini's multiple plume flythroughs revealed:

- **Salts:** $NaCl$, $NaHCO_3$: direct samples of the subsurface ocean
- **Silica nanoparticles:** imply ocean-floor temperature >90°C (hydrothermal)
- **Molecular hydrogen** (H_2): produced by water–rock serpentinisation
- **Organic molecules:** CO_2, CH_4, NH_3, HCN, complex species up to C_6
- Phosphate detected in 2023 (Postberg et al.), last major missing bioessential element

Enceladus' ocean has all the ingredients (liquid water, energy, organics, phosphorus) for chemoautotrophic life, if it exists there.

Ocean Worlds: Europa & Titan

- **Europa:** Global ocean ~100 km deep, ice shell ~10–30 km
  Maintained by Laplace resonance with Io and Ganymede
  Europa Clipper mission (launched 2024) will investigate habitability
- **Titan:** Subsurface water–ammonia ocean
  Evidence from rotational state (Cassini)
  Dragonfly mission (launch ~2028, arrival ~2034) will explore surface chemistry

These ocean worlds are among the most promising targets for extraterrestrial life: they have liquid water, chemical energy, and organic building blocks.

6. Recent Advances & Outlook

Io’s Tvashtar plume. Credit: NASA/JHUAPL/SwRI

Recent Advances

- **InSight mission** (2018–2022): first seismology on Mars
  Core radius ~1830 km; mantle structure; crustal thickness
  Heat flow probe encountered unexpected soil properties
- **Tidal heating models:** Updated rheological models suggest dissipation in Enceladus may be concentrated in the ice shell
- **Radiogenic budget:** Updated meteorite analyses refine early solar system heating

InSight: Mars Heat Flow and Interior

InSight's seismometer (SEIS) and heat flow probe (HP^3) transformed our view of Mars.

- Over 1300 marsquakes detected during the mission
- Core radius: 1830 ± 40 km, larger than pre-mission estimates
- Partially molten outer core (required by seismic attenuation)
- Crustal thickness: 24–72 km (two-layer model)
- Heat flow probe stuck at 40 cm depth in unexpectedly cohesive regolith
- Upper limit on near-surface heat flux: ~22 mW m^{-2}

Source: Stähler et al. (2021); Khan et al. (2021); Spohn et al. (2022), *Adv. Space Res.*

Europa Clipper: Targeting the Ice Shell

NASA Europa Clipper launched October 2024, arrives at Jupiter in 2030.

- Will make ~50 close flybys of Europa over ~4 years
- **Ice Penetrating Radar (REASON):** maps ice shell thickness and near-surface layers
- **Magnetometer:** constrains ocean salinity from induced response
- **MASPEX + SUDA:** samples the composition of possible plume material
- Thermal mapping will locate regions of thinner ice and recent activity
- Will directly test whether tidal dissipation is concentrated in the shell or in the rocky core

Source: Howell & Pappalardo (2020), *Nature Communications*

Modelling Mars' Thermal Evolution

Mars' thermal history explains key observations:

- Early intense volcanism (Tharsis) → hot magma ocean + dynamo (4.1 Ga)
- Dynamo shutdown ~4.1 Ga → core cooling slowed, crustal magnetisation frozen
- Olympus Mons still eruptive until ~100–200 Ma
- Present-day heat flow ~15–25 mW/m^2 (InSight constraints)
- Partially molten upper core retains heat; slow mantle convection prolongs cooling

Mars' thermal history ties directly to its atmospheric evolution: the loss of the dynamo caused the loss of the field, and eventually the atmosphere.

Io in the Decade of Juno

Juno's extended mission (2021–) has delivered the closest-ever Io observations:

- Global heat flow map: polar regions are hotter than expected, contradicting simple tidal models
- Thermal emission imaging from JIRAM instrument
- Gravity data consistent with a global magma ocean just beneath the crust (Park et al. 2024)
- New implications for the structure of Io's interior: is the "mantle" largely molten?
- Future: IVO (Io Volcano Observer) concept proposal for dedicated Io mission

Source: Park et al. (2024); Keszthelyi et al. (2022)

JUICE: The Jupiter Icy Moons Explorer

ESA's JUICE launched in April 2023, arrives Jupiter 2031.

- Multiple flybys of Europa, Callisto, and Ganymede
- Orbit insertion at **Ganymede** in 2034: first orbiter of a moon other than our own
- Will determine the thickness of Ganymede's ice shell and ocean via laser altimetry and induced magnetic field
- Direct measurement of tidal heating through tidal Love number k_2
- Complementary to Europa Clipper: two flagship ocean world missions in the 2030s

Source: Grasset et al. (2013), *Planet. Space Sci.*

Are Ocean Worlds Habitable?

- Require **three things**: liquid water, energy, chemical nutrients
- **Liquid water:** tidal heating + ice shell insulation
- **Energy:** chemical disequilibria (serpentinisation produces H_2)
- **Nutrients:** rock–water reactions deliver P, S, trace metals
- **Open questions:** redox state, temperature range, ocean pH, mixing timescales
- Ganymede, Europa, Enceladus, Titan, Triton, Pluto and maybe Ceres all have possible subsurface oceans

Source: Nimmo & Pappalardo (2016), *J. Geophys. Res. Planets*

Summary

1. Four heat sources: accretional, gravitational differentiation, radioactive decay, tidal heating
2. Conduction is diffusive ($\tau \sim L^2/\kappa$), too slow for planet-sized bodies
3. Convection dominates in large bodies: $\mathrm{Ra} \gg \mathrm{Ra}_c$
4. Earth: mobile lid (plate tectonics); other rocky bodies: stagnant lid
5. Tidal heating powers Io's volcanism and maintains subsurface oceans on Enceladus, Europa, and Titan
6. Earth's total heat loss: 47 TW (40% radiogenic, 60% primordial)

Questions?

Next: **Lecture 4. Chemical Differentiation & Magnetospheres**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`